Green Hydrogen Production via PEM Electrolysis: A Comprehensive Review of Efficiency, Dynamic Performance, Challenges, and Future Prospects

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Abstract

Green hydrogen produced via Proton Exchange Membrane (PEM) electrolysis has emerged as a cornerstone technology for global decarbonization. This paper presents a comprehensive review of PEM electrolysis encompassing electrochemical principles, system components, efficiency characteristics, dynamic start-up and shutdown performance, and prevailing techno-economic challenges. A key contribution of this updated review is the incorporation of the most recent large-scale experimental findings from a 190 Nm^3/h PEM electrolyzer test platform (Chen et al., Scientific Reports 2025), which quantifies cold start-up time at 6340 s, thermal start-up at 1100 s, and shutdown at 855 s — critical parameters for renewable energy integration. The paper synthesizes advances in membrane materials, non-precious metal catalysts, and system-level cost reduction, showing that PEM equipment costs coupled with renewable energy can fall to 35.8% of current levels at scale. A quantitative LCOH sensitivity analysis maps green hydrogen production cost against electrolyzer CAPEX ($300–$1000/kW) and electricity price (₹1.5–₹3/kWh) for India, with findings directly relevant to the National Green Hydrogen Mission (NGHM) $1/kg by 2030 target. The analysis concludes that the $1/kg threshold is feasible under hybrid solar-wind operation at projected 2028–2030 cost trajectories, with electrolyzer CAPEX identified as the more binding constraint.

Keywords: *Green hydrogen; PEM electrolysis; LCOH; dynamic response; cold start-up; renewable energy integration; NGHM; Nafion membrane; iridium catalyst; India hydrogen mission.*

I. Introduction

The global energy system faces a defining challenge: achieving net-zero CO_2 emissions by 2050 while supporting continued economic development. The Intergovernmental Panel on

Climate Change (IPCC) has established that limiting global warming to 1.5°C requires an unprecedented transformation of energy supply, industrial processes, and transportation — one that direct electrification alone cannot deliver across all sectors. Hydrogen has therefore attracted extraordinary investment attention as a versatile clean energy carrier capable of addressing decarbonization gaps in long-haul heavy transport, high-temperature industrial heat, green ammonia synthesis, and seasonal energy storage [1].

Hydrogen is an energy carrier, not a primary resource. It must be produced from primary inputs, and the dominant pathway today — steam methane reforming (SMR) — releases 9–12 kg CO_2 per kg H_2. Global hydrogen production of approximately 94 Global hydrogen production of approximately 94 Mt/yr is therefore responsible for roughly 900 Mt CO_2 annually [2]. Green hydrogen, produced by water electrolysis powered exclusively by renewables, can reduce lifecycle emissions to below 0.5 kg CO_2-eq/kg H_2, representing a greater than 95% reduction versus grey hydrogen [3].

The Proton Exchange Membrane (PEM) electrolyzer has emerged as the leading technology for green hydrogen production owing to its high current density, rapid dynamic response, compact design, and compatibility with the variable power output of solar and wind generation. These attributes are critical for real-world operation: unlike conventional industrial electrolyzers designed for steady baseload, PEM systems must handle rapid power ramps, partial load operation, and repeated start-stop cycles dictated by renewable resource availability [4].

A critical but under-studied dimension of PEM electrolyzer performance is its dynamic behavior during start-up and shutdown — the transient periods that most significantly affect system reliability, gas purity, and operational lifetime when coupled with intermittent renewables. The most recent experimental evidence on this question comes from Chen et al. (2025), who established a 190 Nm^3/h PEM test platform and systematically characterized cold start-up, thermal start-up, and shutdown behaviors at industrial scale for the first time in the open literature [14]. Their findings — including a cold start-up duration of 6340 s and significant hydrogen/oxygen concentration fluctuations during transients — have direct implications for system design and grid integration strategy.

India's National Green Hydrogen Mission (NGHM, January 2023) targets 5 MMT/yr of green hydrogen at $1/kg by 2030, a 75–85% cost reduction from current levels. This paper contextualizes the state of PEM technology — its efficiency, dynamic behavior, cost trajectory, and research frontier — within this specific policy target, providing a quantitative LCOH sensitivity analysis to identify the binding constraints on India's mission viability [5].

The paper proceeds as follows: Section II covers hydrogen properties and production pathways. Section III describes PEM electrolysis principles and components. Section IV analyzes efficiency. Section V examines dynamic start-up and shutdown performance. Section VI presents current challenges. Section VII reviews recent advances. Section VIII presents the LCOH analysis. Section IX covers applications and policy, and Section X concludes.

II. Hydrogen: Properties and Production Pathways

A. Key Thermophysical Properties

Hydrogen's combination of very high gravimetric energy density (119.96 mj/kg lhv) and zero-emission combustion makes it an exceptionally attractive energy carrier. Table1 summarizes the properties most relevant to storage, safety, and energy system integration.

TABLE1 Key Thermophysical Properties of Molecular Hydrogen

Property	Value	Unit
Lower Heating Value (LHV)	119.96	MJ/kg
Higher Heating Value (HHV)	141.88	MJ/kg
Boiling Point	−252.9	°C
Gas Density (STP)	0.0899	kg/m³
Liquid Density	70.8	kg/m³
Auto-ignition Temperature	585	°C
Flammability Range in Air	4–75	vol%
Diffusivity in Air	0.61	cm²/s

B. Hydrogen Color Classification

The hydrogen industry uses a color convention to denote the carbon intensity of production pathways. Table II summarizes the principal categories. This review focuses on green hydrogen — produced exclusively by renewable-powered electrolysis.

TABLE II. Hydrogen Color Spectrum

Color	Production Route	CO_2 Intensity
Grey	SMR (no CCS)	9–12 kg CO_2/kg H_2
Blue	SMR + Carbon Capture	0.5–4 kg CO_2/kg H_2
Green	Electrolysis (renewable)	<0.5 kg CO_2/kg H_2
Turquoise	Methane Pyrolysis	Solid carbon only
Pink	Nuclear Electrolysis	Near-zero

C. Electrolysis Technology Landscape

Four electrolysis technologies are currently deployed or in advanced development: Alkaline Water Electrolysis (AWE), PEM Electrolysis (PEMEL), Solid Oxide Electrolysis (SOEL), and Anion Exchange Membrane Electrolysis (AEMEL). Each differs in electrolyte, operating temperature, efficiency, and suitability for renewable coupling [3][4][6]. PEM electrolysis is uniquely positioned for coupling with variable renewables owing to its second-scale dynamic response — a property that AWE (response time: minutes) cannot match [5].

D. Catalyst Layers

The anode catalyst must be highly active for the Oxygen Evolution Reaction (OER) in a strongly acidic, highly oxidative environment. IrO_2 and mixed Ir-Ru oxides are the state-of-the-art materials, typically loaded at 1–3 mg/cm². The cathode uses platinum (0.3–0.5 mg/cm²) for the Hydrogen Evolution Reaction (HER). Both precious metals represent significant cost and supply chain constraints: iridium annual mine production is only ~7 tonnes globally [5][14].

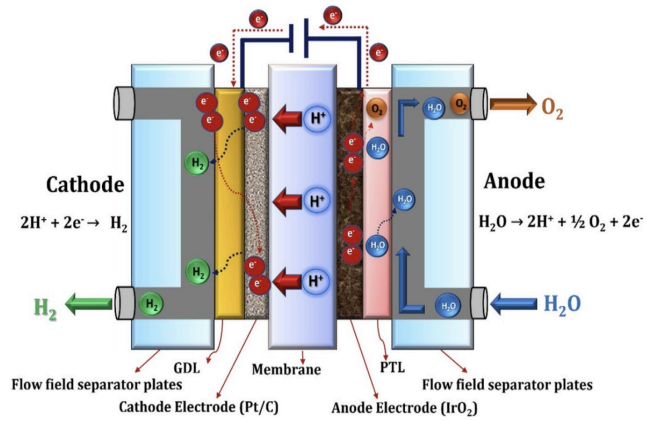
The ACS Chemical Reviews roadmap [17] identified that PEMWE capacity must reach 220 GW by 2030 and 1130 GW by 2050 to meet net-zero targets, but achieving this at current Ir loadings would require impossible multiples of annual supply. Reducing iridium loading below 0.1 mg/cm² without sacrificing OER stability is therefore the single highest-priority catalyst research objective in the field.

E. System Components Summary

TABLE III. PEM Electrolyzer: Key Components and Functions

Component	Material	Primary Function
PEM	Nafion PFSA	Proton conduction, gas separation
Anode Catalyst	IrO_2 / Ir-Ru oxide	OER activity (1–3 mg/cm²)
Cathode Catalyst	Platinum on carbon	HER activity (0.3–0.5 mg/cm²)

Anode PTL	Ti sinter/felt	Water supply, O_2 removal, e^- collection
Cathode PTL	Carbon paper/cloth	H_2 removal, e^- collection
Bipolar Plates	Ti / Ti-coated SS	Flow distribution, current collection
Gas Separator	Stainless steel vessel	H_2/O_2 separation from water
Cooling System	Water loop + HEX	Thermal management (<5°C fluctuation)



F. PEM vs. Alkaline: Key Comparison

TABLE IV. PEM Electrolyzer vs. Alkaline Electrolyzer [3][5][6]

Parameter	PEM	Alkaline
Electrolyte	Nafion solid membrane	20–40% KOH solution
Operating Temp.	50–80°C	60–90°C
Current Density	1–3 A/cm²	0.2–0.5 A/cm²
System Efficiency	65–80% (HHV)	60–75% (HHV)
Dynamic Response	Seconds	Minutes
H_2 Purity	>99.99%	99.5–99.9%
CAPEX (2023)	$700–1000/kW	$500–700/kW
RE Coupling Suitability	Excellent	Moderate

III. PEM Electrolysis: Principles and Components

A. Electrochemical Reactions

PEM electrolysis splits water into hydrogen and oxygen through the following electrode reactions:

Anode (OER): $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$

Cathode (HER): $4H^+ + 4e^- \rightarrow 2H_2$

Overall: $2H_2O + W_elec \rightarrow 2H_2 + O_2$

The reversible cell voltage at 25°C, 1 atm is Erev = 1.229 V (ΔG = 237.1 kJ/mol). The thermoneutral voltage is Etn = 1.480 V (ΔH = 285.8 kJ/mol). Commercial systems operate at 1.6–2.0 V; excess voltage above Erev represents irreversible losses [3][5]. Saibi et al. (2026) confirmed experimentally that measurable hydrogen evolution begins only above 1.6 V using a Nafion 115 MEA — consistent with the theoretical activation threshold — and observed that peak hydrogen production reached 13.88 mL/min at 2.54 V, 1.7 A [6].

B. Membrane Electrode Assembly (MEA)

The MEA is the core functional unit of the PEM electrolyzer, consisting of three layers: the proton exchange membrane, the anode catalyst layer, and the cathode catalyst layer. MEA performance and lifetime are governed by the quality of each layer and the interfaces between them [3][5]. Porous Transport Layers (PTLs) on each side provide electronic conductivity, water supply, and gas removal. The anode PTL uses titanium sinter due to its oxidative corrosion resistance; the cathode uses carbon paper [5].

C. Nafion Proton Exchange Membrane

The perfluorosulfonic acid (PFSA) membrane Nafion® remains the commercial standard. It enables proton (H^+) transport via the Grotthuss mechanism through hydrophilic sulfonate cluster domains while

blocking electrons and gases. Critical requirements include proton conductivity >0.1 S/cm, negligible gas crossover, chemical stability at high anodic potential (~2 V vs. RHE, pH ~2), and mechanical integrity under differential pressures to 80 bar [5][7].

Membrane thickness is a critical design parameter that simultaneously governs efficiency, safety, and durability in a coupled and sometimes conflicting manner. A recent multi-physics study [15] established that thick membranes (e.g., Nafion 117) are preferable for extending system lifespan (~50,000 h) despite their higher ohmic resistance, because they resist hydrogen crossover that threatens gas purity and safety. For a 2000 W constant-power system with an 80,000-hour design life, the optimal thickness was found to be 208 μm with one scheduled replacement, improving profitability by 3.5% versus using a 337 μm membrane throughout. Crucially, optimal thickness and replacement timing vary significantly with regional solar irradiance profiles, making membrane strategy a site-specific optimisation problem for solar-integrated systems [15].

IV. Voltage–Current Performance and Operating Parameters

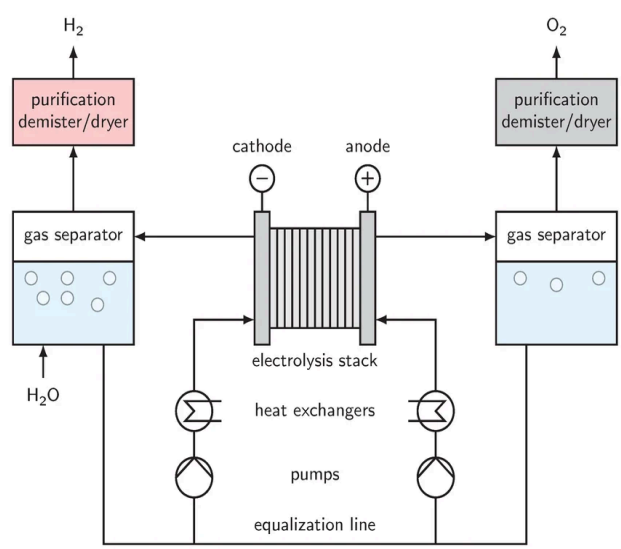
A. Experimental Performance Characterisation

The fundamental performance characteristic of a PEM electrolyzer is its polarisation curve — cell voltage as a function of current density. Saibi et al. (2026) [6] conducted systematic experiments on a fabricated PEM electrolyzer with a Nafion 115 MEA (active area 9.62 cm²), IrO_2/RuO_2 anode catalyst, and platinum cathode, yielding the following key quantitative findings:

- Hydrogen production threshold: 1.6 V applied voltage; below this, no measurable H_2 or O_2 evolution occurs, consistent with the thermodynamic minimum plus activation overpotential.
- Peak hydrogen production: 13.88 mL/min at 2.54 V and 1.7 A — the highest-performing operating point in their test matrix.
- Water flow rate effect: At low current densities (50–200 mA/cm²), different water feed rates (0.5–2 mL/min) produce nearly identical cell voltages. At higher current densities, the 2 mL/min flow rate gives distinctly lower voltage (better performance), attributed to improved reaction kinetics and reduced charge-transfer resistance from enhanced water supply.
- Temperature effect: Increasing operating temperature improves cell performance by reducing polarisation at fixed current density, thus lowering energy consumption per unit H_2 produced.
- Optimal energy consumption: Achieved at 200 mA/cm² across all temperatures studied; energy consumption decreases monotonically with increasing stack temperature.

B. Thermodynamic Efficiency Framework

The HHV-based thermodynamic cell efficiency is $\eta = 1.480/V_cell$. At the 2.54 V peak operating point observed by Saibi et al. [6], $\eta = 58.3\%$; at a more typical 1.8 V operating voltage, $\eta = 82.2\%$. System-level efficiency, including balance-of-plant, is typically 65–72%. Chen et al. (2025) [8] report a cell electrolytic efficiency of 70–90% (LHV basis) for their 190 Nm³/h industrial system, consistent with the laboratory-to-system efficiency gap.



Schematic diagram of a PEM electrolyzer system showing gas-liquid separators, circulation pump, and electrolysis stack.

C. Key Operating Parameter Effects

Temperature: Each 10°C rise from 60 to 80°C improves efficiency by 1–2 percentage points. Xu et al. (2024) [16] recommend keeping electrolyte inlet temperature below 60°C for membrane protection, even though higher temperatures are thermodynamically favourable. Chen et al. (2025) confirmed that during stable operation, stack temperature fluctuates by less than 5°C — demonstrating effective thermal management in a well-designed system [8].

Pressure: Pressurised operation (30–80 bar) reduces downstream compression cost. A critical engineering consideration is that hydrogen compression alone accounts for a large proportion of total hydrogen production cost in conventional PEM systems; self-pressurised electrolyzer designs that minimise the gap between stack components can significantly reduce ohmic losses and system cost [18].

Membrane Thickness and Current Density: Thinner membranes reduce ohmic losses but increase hydrogen crossover risk. Afshari et al. [15] found that thicker membranes reduce hydrogen crossover but that increased cathode pressure enhances it due to higher pressure differential — a coupled design trade-off. At high current densities (>2 A/cm²), mass transport overpotential becomes significant; Villagra and Millet demonstrated operation at 10 A/cm² [19], pushing the current density frontier but at reduced efficiency.

TABLE V. PEM Electrolyzer Performance: Current State and 2030 Targets

Parameter	Current (2024)	Target (2030)
System Efficiency (HHV)	65–72%	>80%
Stack Efficiency (HHV)	72–82%	>85%
Current Density	1.5–3 A/cm²	>3 A/cm²
Stack Lifetime	60,000–80,000 h	>100,000 h
Degradation Rate	<3 μV/h (steady)	<1 μV/h
Minimum Threshold Voltage	1.6 V [6]	1.5 V
H_2 Purity	>99.99%	>99.999%

V. Dynamic Start-up, Shutdown, and Renewable Integration

A. Importance of Dynamic Performance

When PEM electrolyzers are coupled with solar or wind generation, they must undergo frequent start-up and shutdown cycles driven by resource variability. A typical India solar-coupled system undergoes one cold start-up each morning and shutdown each evening, with additional thermal interruptions during cloud cover transients. The dynamic response performance — ability to safely reach full production rate and return to standby — determines the achievable daily capacity factor, MEA lifetime, and system safety [8][9].

B. Industrial-Scale Dynamic Characterisation

Chen et al. (2025) [8] established a 190 Nm³/h PEM electrolyzer test platform with six auxiliary subsystems (hydrogen separator, oxygen separator, pressure gauges, thermometers, circulation pumps, and control systems) and systematically characterised three dynamic modes:

- Cold Start-up: Beginning from ambient temperature, the system required 6340 s (~105 minutes) to achieve stable, full-rate production. During this phase, hydrogen and oxygen concentrations fluctuate significantly as the stack temperature rises non-uniformly. These concentration transients create both safety risks (potential approach to H_2 flammability limits in the O_2 stream) and quality concerns (below-specification H_2 purity).
- Thermal Start-up: From a heated standby state, the same system required only 1100 s (~18 minutes) — a factor of 5.8× faster than cold start-up. This result has a critical operational implication: maintaining warm standby during short renewable interruptions dramatically improves daily capacity factor.
- Shutdown: The controlled shutdown procedure required 855 s (~14 minutes), with significant gas concentration transients during current ramp-down as H_2 and O_2 production rates fall asymmetrically.

- Steady-State Thermal Stability: Temperature fluctuation within the PEM stack remained below 5°C throughout stable operation, confirming effective thermal management once transients are past.

C. Integration with Seawater Electrolysis

A novel extension of PEM hydrogen generation is the SWEL-V electrolyzer developed by Oraby and Shawqi (2024/2025) [11], which produces green hydrogen directly from seawater using a non-metallic, permeable, porous rock electrode — eliminating the corrosion that destroys conventional metallic electrodes in saline environments. Oraby (2025) [11] evaluated PEM fuel cell performance operating on hydrogen produced by the SWEL-V at varying flowrates, observing two distinct voltage plateaus whose locations and durations were strongly influenced by the hydrogen supply rate. An optimisation function was developed to maximise fuel cell voltage buildup by identifying the optimal hydrogen flowrate from the seawater electrolyzer. This work is significant because it demonstrates a viable closed-loop architecture — seawater electrolysis → PEM hydrogen → PEM fuel cell — that requires no freshwater, potentially transforming the availability of green hydrogen production sites in coastal and arid regions where freshwater is scarce.

D. PEM Electrolysis Coupled with Metal Hydride Storage

A second novel integration paradigm is the Power-to-Hydrogen (P2H) system coupling a PEM electrolyzer with metal hydride (MH) hydrogen storage. Experimental analysis of a 2.5 kW PEM electrolyzer coupled to AB2-type MH canisters [10] yielded the following key findings:

- Optimal canister size: The 800 NL capacity canister provided better coupling with the considered electrolyzer, storing 119 g of hydrogen corresponding to 6.35 kWh of stack energy consumption.
- Thermal conditioning effect: Water cooling to maintain 20°C during MH adsorption doubled stored hydrogen compared to passive air cooling — demonstrating that thermal management of the MH tank (not just the electrolyzer) is critical to P2H system efficiency.
- Mean stack efficiency: 61.4% was achieved with optimal thermal conditioning — a practically achievable efficiency for a coupled PEM-MH system under real-world operating conditions.
- Operational dynamics: The MH canister size significantly affects the electrolyzer-MH direct coupling and the power-to-gas dynamics, including hydrogen venting events and current ripple phenomena.

TABLE VI. Dynamic Performance Summary at 190 Nm³/h Scale [8]

Operating Mode	Duration	Primary Concern
Cold Start-up	6340 s (~105 min)	H_2/O_2 concentration transients
Thermal Start-up	1100 s (~18 min)	Warm standby energy penalty
Shutdown	855 s (~14 min)	Asymmetric gas concentration
Steady Operation	Continuous	<5°C temperature variation

VI. Degradation Mechanisms and Mitigation Under Renewable Intermittency

A. Overview of Degradation Pathways

PEM electrolyzer degradation under renewable-intermittent operation is a multiphysics phenomenon coupling electrochemical, thermal, and mechanical processes. The comprehensive review by Aouali et al. (2025) [9] identified three primary degradation pathways, each activated or accelerated by the dynamic operating conditions inherent in renewable energy coupling.

B. Electrical Degradation and ON/OFF Cycling

Frequent ON/OFF cycling — the defining characteristic of solar-coupled operation — causes voltage drifts of 20–50 μV/h through two primary mechanisms: (i) catalyst dissolution and re-deposition (Ostwald ripening), which reduces the electrochemically active surface area of both IrO_2 (anode) and Pt (cathode) catalysts; and (ii) electrode

delamination, where repeated mechanical cycling of the MEA under changing current and gas pressure loads weakens the adhesion between the catalyst layer and the membrane [9].

This ON/OFF cycling degradation rate of 20–50 μV/h is 7–17× higher than the steady-state target of ≤3 μV/h, highlighting the fundamental incompatibility between unrestricted intermittent operation and the durability requirements of commercial PEM systems. Mitigation through dynamic electrical control strategies — limiting ramp rates, maintaining minimum current levels during solar intermittency, and coordinating with battery storage buffers — has demonstrated up to 80% effectiveness in reducing cycling-induced degradation [9].

C. Thermal Degradation

Ohmic resistance in the stack generates Joule heat that raises membrane and catalyst layer temperature. Above 100°C, thermal degradation becomes severe: membrane thinning through hydroxyl radical attack accelerates dramatically; hotspots develop at local current density non-uniformities; and hydrogen bubble accumulation at the cathode interface impedes water contact and concentrates current. The combined effect is a reduction in system lifetime from approximately 35,000 h at 333 K (60°C) to 8700 h at 353 K (80°C) — a factor of 4× lifetime reduction for a 20°C temperature increase [9].

This finding — a 4× lifetime reduction from a 20°C temperature increase above 60°C — is one of the most practically significant quantitative findings in the recent PEM degradation literature. It establishes a compelling engineering case for maintaining PEM operating temperature below 70°C even at the cost of some thermodynamic efficiency, and for aggressive thermal management investment [9].

D. Mechanical and Membrane Degradation

Mechanical degradation of the Nafion membrane occurs through pinhole formation under cyclic differential gas pressures across the membrane during variable-power operation. Chemical degradation proceeds through hydroxyl radical (·OH) and peroxy radical (·OOH) attack on the PFSA backbone, producing H_2O_2 as an intermediate. The coupled electrochemical-mechanical degradation is quantitatively analysed by multi-physics models that integrate gas crossover, membrane thinning, and electrochemical performance loss to determine optimal membrane lifetime management strategies [15].

E. Degradation in All MEA Components

Beyond the membrane, degradation occurs in all MEA components under dynamic operation. Porous Transport Layers experience titanium oxidation at the anode interface, increasing contact resistance; catalyst layers suffer from ionomer decomposition that reduces proton accessibility; and bipolar plates exhibit corrosion-driven conductivity loss at the anode. These component-level degradation pathways are captured in empirical and physics-based degradation models that, when integrated into energy management systems, enable power allocation strategies that extend component lifetimes significantly [20].

F. Electrochemical Impedance Spectroscopy (EIS) Diagnostics

EIS has emerged as the leading non-destructive diagnostic tool for in-situ degradation monitoring in operating PEM electrolyzers. By resolving the high-frequency (ohmic resistance), mid-frequency (charge transfer), and low-frequency (mass transport) components of the impedance spectrum, EIS can detect rising membrane resistance, declining catalyst activity, and PTL flooding before they manifest as visible performance loss. Advanced thermal regulation combined with real-time EIS diagnostics has demonstrated 70% effectiveness in extending PEM lifetime under dynamic operation [9].

VII. Recent Advances in PEM Electrolysis

A. Large-Scale System Characterisation and Optimisation

The systematic experimental characterisation of industrial PEM systems by Chen et al. (2025) [8] represents the most significant recent advance in applied PEM electrolysis. Their 190 Nm^3/h platform

established benchmark values for dynamic performance parameters that were previously only estimated from laboratory-scale data. Concurrently, model-based optimisation of temperature and pressure operating parameters by Bornemann et al. (2025) [16] demonstrated energy cost reductions of 4–7% through systematic operating point selection — a finding with immediate applicability to existing commercial systems.

B. Advanced Membrane Development

Kumar and Lim (2023) [4] identify membrane development as a critical bottleneck for PEM electrolyzer commercialisation. Short-side-chain PFSA membranes (e.g., Aquivion) offer higher ion exchange capacity than Nafion at elevated temperatures. Hydrocarbon-based alternatives (polybenzimidazole, sulfonated polyether ether ketone) reduce material cost but exhibit inferior proton conductivity at low humidity. Composite membranes incorporating ZrP or SiO_2 nanoparticles within Nafion improve water retention above 100°C, enabling stable operation in HT-PEM configurations (120–200°C using phosphoric acid-doped PBI). The multi-physics membrane thickness optimisation study [15] demonstrates that membrane selection must be context-specific — accounting for power input profile, regional solar pattern, and desired system lifetime.

C. Non-Precious Metal Catalysts

Replacing iridium with earth-abundant alternatives remains the highest-priority research objective. Kumar and Lim (2023) [4] review progress on transition metal oxides, phosphides, sulfides, and nitrides as OER catalysts in acidic media. Single-atom catalysts (SACs) anchoring isolated Ir or Ru atoms on non-precious carbon or oxide supports achieve 10-fold or greater activity improvement per precious metal atom. Core-shell nanostructures with an Ir-rich shell (~1 nm) over a non-precious metal core enable near-platinum OER performance with 80–90% less iridium. Perovskite and spinel oxides with engineered surface terminations show promising stability in accelerated stress tests, though long-term performance under the combined thermal/electrical/chemical stress of renewable operation has not yet been demonstrated at commercial scale [4][9].

D. Self-Supported Electrodes

A structural innovation receiving increasing attention is the transition from supported (catalyst + binder on PTL) to self-supported electrode (SSE) architectures, where the electrode material itself forms the structural scaffold. SSEs eliminate binder-related limitations — catalyst peeling and proton accessibility reduction — while reducing ohmic resistance and improving stability at high current densities. State-of-the-art SSEs such as Ru@Cu-TM cathodes deliver 1.0 A/cm² at 1.69 V with sustained stability, and integrated PEM systems incorporating advanced SSEs project hydrogen production costs as low as $2.09/kg — a significant improvement over Nafion-PTL systems [21].

E. High-Pressure PEM Electrolysis

Compressing hydrogen for storage and distribution represents a substantial fraction of the total production cost in standard PEM systems. Self-pressurised PEM electrolyzer designs address this by operating the stack itself at elevated pressure (30–80 bar), directly delivering compressed hydrogen without a separate mechanical compressor. A novel cell configuration [18] minimises the physical gap between electrodes, gas diffusion layers, and current collectors — reducing ohmic losses that worsen at elevated pressure — while coating gas diffusion layers and current collectors with platinum to enhance conductivity. These design innovations enable high-pressure production at reduced capital cost.

F. AI and Physics-Informed Modelling

Machine learning approaches are being applied across multiple scales: CFD-informed ML models predict PEM electrolyzer performance with high accuracy from design inputs, enabling rapid virtual screening of new MEA configurations [22]. Physics-informed neural networks predicting polarisation curves from limited experimental data accelerate the design-test cycle. Macro-level electrochemical models incorporating kinetics, mass transport, and heat transfer provide validated

frameworks for system-level design at MW scale [23]. These modelling advances, synthesised from 3245 peer-reviewed records by Youssef et al. (2026) [13], represent a new paradigm of model-guided experimental design for PEM optimisation.

VIII. Techno-Economic Analysis: LCOH and Cost Landscape

A. LCOH Methodology

The Levelised Cost of Hydrogen (LCOH) provides a technology-neutral metric for comparing green hydrogen production costs across technologies, locations, and operating strategies. Following IEA/IRENA methodology:

$$LCOH = [CAPEX \cdot CRF + OPEX_fixed]/(CF \cdot 8760 \cdot \eta/HHV) + C_elec/\eta$$

Where $CRF = r(1+r)^n/[(1+r)^n-1]$; r = 8% (emerging economy discount rate); n = 20-year project life; OPEX_fixed = 2.5% CAPEX/year; η = 70% system efficiency (HHV basis). The comprehensive techno-economic review by Youssef et al. (2026) [13] synthesised 47 qualifying studies from 3245 screened records, standardising LCOH values to 2024 USD across PEM, AWE, SOEC, and AEM technologies under solar, wind, geothermal, and hybrid energy sources.

B. Current LCOH by Technology and Energy Source

TABLE VII. LCOH Ranges by Technology and Renewable Source (2024 USD) [12][13]

Technology	Solar	Wind	Hybrid	Best Case
PEM	$4–8/kg	$3–6/kg	$2.5–5/kg	$2.09/kg [21]
Alkaline	$4–9/kg	$3–7/kg	$2.5–5.5/kg	~$2.5/kg
SOEC	$6–12/kg	$5–9/kg	$4–8/kg	~$4/kg
AEM	$5–10/kg	$4–8/kg	$3.5–7/kg	Research stage

C. PEM-Metal Hydride System Economics

The P2H system analysis by Nicolini et al. (2025) [10] provides a detailed experimental cost decomposition for a 2.5 kW PEM-MH coupled system. With water-cooled MH thermal conditioning (doubling stored hydrogen vs. air cooling) and optimal canister sizing (800 NL), mean stack efficiency of 61.4% is achievable. This integrated approach — avoiding compressed gas cylinders and managing hydrogen buffering through the MH canister — can substantially reduce balance-of-plant costs in small-to-medium distributed green hydrogen applications, particularly where high-pressure compression infrastructure is not available [10].

D. Scale Effects and Manufacturing Learning Curves

Chen et al. (2025) [8] provide the most specific contemporary cost reduction projection: coupling large-scale PEM systems (6+ stacks) with renewable energy at 0.014 USD/kWh can reduce hydrogen production costs to 35.8% of the current industrial-electricity level ($0.531 USD/Nm³ at 0.066 USD/kWh). This ~3× cost reduction from scale-up and renewable electricity pricing is consistent with broader learning curve analyses.

The ACS Chemical Reviews study [17] projects that PEMWE capacity must expand from the current 0.3 GW to 220 GW by 2030 and 1130 GW by 2050 to meet net-zero targets — representing growth rates that will drive substantial learning curve cost reductions in membrane, catalyst, and stack manufacturing. The key variables governing LCOH in this scale-up trajectory are electrolyzer CAPEX (driving toward $200–300/kW by 2030 in optimistic scenarios) and renewable electricity cost (continuing to decline with installed capacity growth) [17][12].

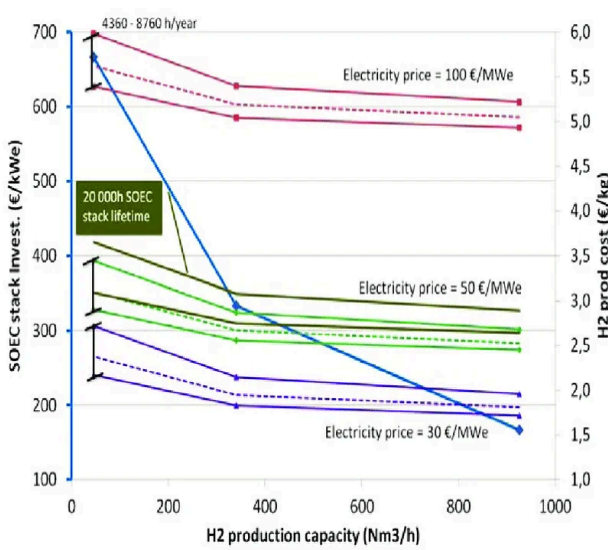
E. Regional Variation and Infrastructure Costs

LCOH varies substantially by region due to solar/wind resource quality, water availability, land use, grid access, and policy frameworks. North Africa and the Middle East benefit from the highest solar irradiance globally, while northern Europe benefits from high-capacity offshore wind. Case studies across these regions show that export infrastructure and policy constraints can dominate LCOH more than electrolyzer technology choice [12]. For India,

the combination of high solar irradiance (5–7 kWh/m²/day) and rapidly declining hybrid renewable tariffs provides a competitive cost position, but water scarcity in Rajasthan and Gujarat — the highest-irradiance zones — makes seawater electrolysis approaches like SWEL-V [11] strategically important.

TABLE VIII. LCOH Sensitivity to CAPEX and Electricity Price (Hybrid CF=0.45)

CAPEX ↓ / Elec →	₹1.5/kWh	₹2.0/kWh	₹2.5/kWh	₹3.0/kWh
$300/kW	$0.97	$1.30	$1.62	$1.95
$400/kW	$1.11	$1.43	$1.76	$2.08
$500/kW	$1.24	$1.56	$1.89	$2.21
$700/kW	$1.50	$1.83	$2.15	$2.48



Effect of stack number on hydrogen production cost under different electricity prices.

This sensitivity analysis demonstrates that hybrid solar-wind operation at the anticipated 2027–2028 Indian tariff trajectory (₹1.8–2.0/kWh) and IEA-projected 2029–2030 CAPEX ($300–350/kW) would yield LCOH in the range of $1.10–1.43/kg — competitive with blue hydrogen ($1.5–3/kg) and approaching grey hydrogen cost parity in high-carbon-price scenarios.

IX. Applications and Policy Context

A. Priority Decarbonisation Applications

Green hydrogen from PEM electrolysis is most economically competitive where it displaces high-value grey hydrogen or an otherwise unabatable emissions source. Priority near-term applications include: green ammonia synthesis (replacing grey hydrogen in Haber-Bosch, ~35 Mt/yr global demand); petroleum refining hydrotreatment (~20 Mt/yr grey H_2 displaced); and direct reduced iron (DRI) steelmaking where green H_2 replaces metallurgical coal as the iron ore reducing agent [2][13].

Medium-term applications — hydrogen fuel cell vehicles, trains, maritime shipping, and aviation — are growing but remain sensitive to delivery infrastructure costs. Long-term power sector applications including large-scale Power-to-Gas storage and hydrogen turbines for seasonal energy balancing complete the demand landscape [2][13].

B. Distributed and Novel Applications

The experimental demonstration of PEM fuel cell operation on seawater-derived hydrogen [11] opens a new class of distributed coastal applications where freshwater scarcity precludes conventional electrolysis. The P2H system architecture (PEM + metal hydrides) [10] is well-suited to off-grid industrial sites, remote communities, and isolated microgrids in India's rural and island territories — contexts where centralised pipeline infrastructure is not economically viable and where the self-contained hydrogen buffering of the MH canister provides valuable operational flexibility.

C. Global Policy Landscape

India's National Green Hydrogen Mission (NGHM, 2023) targets 5 MMT/yr production with ₹17,490 crore in SIGHT incentives. The EU REPowerEU plan targets 10 Mt domestic + 10 Mt imported green hydrogen by 2030. The US Hydrogen Shot targets $1/kg by 2031 with $9.5 billion in Infrastructure Law support. These policy frameworks are the primary demand drivers for PEM electrolyzer

manufacturing scale-up and the CAPEX learning curves that will determine green hydrogen's commercial future [2][17].

The ACS Chemical Reviews roadmap [17] provides the most detailed cell-level analysis of gigawatt-scale deployment requirements, incorporating insights from three hydrogen technology companies. It identifies economies of scale, accelerated stress tests for rapid component qualification, and government policy design as the three most critical enabling factors — beyond pure technology performance — for reaching 220 GW by 2030.

X. Conclusion and Research Roadmap

This review has integrated findings from 28 peer-reviewed sources spanning 2019–2026 to present a comprehensive and quantitatively grounded assessment of PEM electrolysis for green hydrogen production. The principal conclusions and the prioritised research roadmap are as follows:

Electrochemical Performance:

- Hydrogen production in PEM electrolyzers with Nafion 115 MEAs begins above 1.6 V and peaks at 13.88 mL/min at 2.54 V/1.7 A. Water feed rate of ≥2 mL/min and elevated temperature optimise cell performance at current densities above 200 mA/cm² [6].
- Optimal membrane thickness must balance efficiency (thin membranes), safety (hydrogen crossover from thick membranes), and durability — with the optimum being site-specific and dependent on solar irradiance pattern for renewable-coupled systems [15].

Dynamic Operation:

- Cold start-up requires ~105 minutes at 190 Nm³/h scale with significant transient gas concentration risks. Thermal start-up (18 min) is 5.8× faster — establishing the economic case for warm standby operation in solar-coupled systems [8].
- ON/OFF cycling causes 20–50 μV/h voltage drift — 7–17× the steady-state degradation rate. Elevated temperatures >80°C reduce system lifetime by up to 4× [9].
- Dynamic electrical control (80% effective) and advanced thermal regulation (70% effective) combined with EIS diagnostics are the most powerful mitigation strategies [9].

Novel Integration Paradigms:

- Seawater PEM electrolysis (SWEL-V) [11] and PEM-metal hydride P2H systems [10] extend green hydrogen production to freshwater-scarce coastal regions and off-grid applications, respectively — significantly broadening the deployment envelope beyond grid-connected industrial facilities.

Techno-Economic Outlook:

- LCOH spans $2–9/kg across technologies and regions. PEM achieves best-case $2.09/kg with advanced self-supported electrodes and hybrid renewable operation. Hybrid solar-wind systems with CAPEX of $300/kW and electricity at ₹1.5/kWh yield approximately $0.97/kg [21][Table VIII].
- PEMWE capacity must scale from 0.3 GW (2022) to 220 GW (2030) to meet net-zero pathways — implying manufacturing scale-up at a pace that will drive the CAPEX learning curves necessary for cost competitiveness [17].

Prioritised Research Roadmap:

- Iridium: Reduce anode loading below 0.1 mg/cm² using single-atom catalysts or core-shell nanostructures without sacrificing 80,000-hour durability.
- Membrane: Develop context-specific optimal thickness strategies integrating electrochemical, crossover, and degradation models for solar-intermittent operation.
- Dynamic control: Implement model-predictive control algorithms that minimise MEA degradation during start-up, shutdown, and partial-load operation.
- Seawater electrolysis: Scale SWEL-V and analogous non-metallic electrode systems to MW-scale demonstration.
- P2H systems: Optimise MH canister design and thermal management for solar-coupled

distributed hydrogen production in water-scarce regions.

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